

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456353.73176 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.178 +/- 0.012
Transit depth (Rp/Rs)^2	3.16 +/- 0.44 [%]
Orbital Inclination (inc)	82.13 +/- 1.14
Airmass coefficient 1 (a1)	1.122 +/- 0.013
Airmass coefficient 2 (a2)	-0.0796 +/- 0.009
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	None
Optimal Aperture	4.02
Optimal Annulus	13.82
Transit Duration (day)	0.052 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.178 ± 0.012 vs 0.1594 (deviation 1.55σ)
Duration fit vs published	0.052 d vs 0.0512 d (deviation 0.2σ)
Mid-time O–C	0.5 min
Depth SNR	14.8
Residual scatter	1.18 %

Light curve

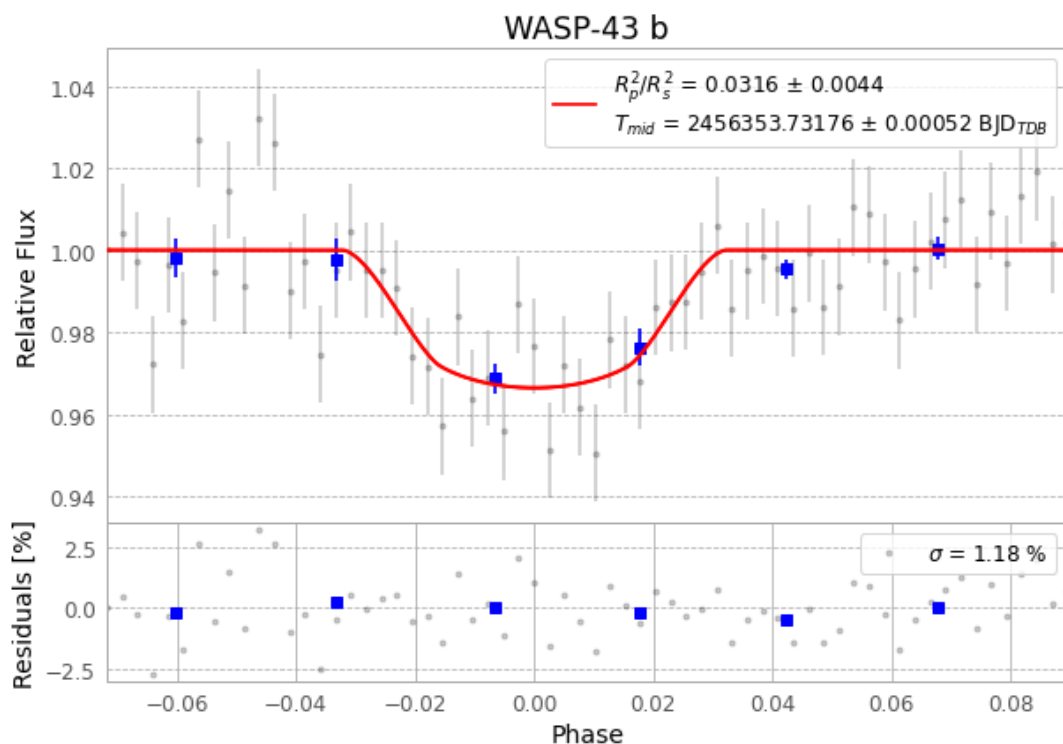


Figure 1 — FinalLightCurve_WASP-43 b_2013-03-01.png (SHA-256 566e07902272135c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d5d3e41bac7f9a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43130302040012.FITS → WASP-43130302070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130302.log (2072289bc782...), FinalLightCurve_WASP-43 b_2013-03-01.png (566e07902272...), FinalLightCurve_WASP-43 b_2013-03-01.csv (7f0e199d5d26...)

Compute resources

Wall time 122.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 23:55Z.